

Determination of Molecular Formula of Potassium Tris(oxalato)ferrate(III) Trihydrate*

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The molecular formula of potassium tris(oxalato)ferrate(III) trihydrate has been derived by analyzing its different chemical components by volumetric analysis and finally using the data in mole ratio method. The experiment presented here is simple, cost-effective, and student-friendly.

Introduction

Determination of the molecular formula of chemical compounds using the mole ratio method both in solid-state and solution is not new. Before the development of modern scientific techniques like X-ray diffraction, these chemical methods were very much effective. Potassium tris(oxalato)ferrate(III) trihydrate (**1**) was first synthesized by J C Bailer Jr. et al., in 1939 [1]. The compound is photoactive and used in chemical actinometry [2]. This article discusses simple redox titrimetric methods used to analyze its different chemical components in percentage amount and using the results to find out their mole ratio, finally determining the molecular formula of the compound. The chemical analysis of the compound has been previously done in two different reports [3, 4]. But, to the best of the author's belief, the method presented in this article is much more simple, cost-effective, and student-friendly. Moreover, none of the earlier communications [3, 4] derives the molecular formula of the compound from the analytical results.

Determination of the molecular formula of chemical compounds using the mole ratio method both in solid-state and solution is not new.

Keywords

Potassium tris(oxalato)ferrate(III) trihydrate, mole ratio method, molecular formula.

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1. Experiments

1.1 Materials

Oxalic acid dihydrate, potassium hydroxide and iron(III) chloride are available commercially.

1.2 Preparation of **1**

1 was synthesized by a modified literature method [5]. 3.0 g (23.8 mmole) oxalic acid dihydrate was taken in 12.5 ml hot water in a 50 ml beaker. To this, 3.8 g potassium hydroxide (67.8 mmole) was added gradually making a clear solution. 2.5 g iron(III) chloride (9.24 mmole) was added to it and stirred continuously until it completely dissolves. After cooling the reaction mixture, it was filtered. A portion of the green filtrate was taken on a porcelain dish and put on a steam bath for solvent evaporation. After a few minutes, the solution was cooled, and parrot green crystals appeared. These were washed with alcohol and water (1:1), and dried by pressing between two filter papers. Yield was ~ 60%.

1.3 Determination of Composition in Percentage

1.3.1 Estimation of Fe(III)

1.0 g of **1** (exactly weighed) was taken in a 250 ml beaker. A minimum volume of concentrated nitric acid was added to it to dissolve the sample completely. The solution was then evaporated to dryness. This process was done in a fume cupboard and repeated thrice minimum. Finally, in the dried beaker, 6N HCl solution was added to dissolve the Fe(III) nitrate. The solution was warmed and titrated against $\frac{N}{10}$ $K_2Cr_2O_7$ following the standard method.

23.5 ml of $\frac{N}{10}$ $K_2Cr_2O_7$ was required to estimate Fe(III).

1000 ml 1N $K_2Cr_2O_7 \equiv 55.85$ g of Fe(III).



$\therefore 1 \text{ ml } \frac{N}{10} \text{K}_2\text{Cr}_2\text{O}_7 \equiv 0.005585 \text{ g of Fe(III).}$

$\therefore 23.5 \text{ ml } \frac{N}{10} \text{K}_2\text{Cr}_2\text{O}_7 \equiv (0.005585 \times 23.5) \text{ g of Fe(III)} = 0.1312 \text{ g of Fe(III).}$

$\therefore \text{Percentage of Fe(III) in the complex} = \frac{0.1312}{1.00} \times 100 = 13.12\%.$

Actual percentage of Fe(III) in the complex = $\frac{55.85}{490.00} \times 100 = 11.39\%.$

1.3.2 Estimation of oxalate

1 is stable with 2(N) sulphuric acid in dark but dissociates quantitatively in light within 5–10 min into Fe(II)oxalate and carbon dioxide [3]. 0.1710 g (exactly weighed) **1** was dissolved in minimum amount of 2 N sulphuric acid and titrated against standardized $0.819 \frac{N}{10} \text{KMnO}_4$.

Volume of $0.819 \frac{N}{10} \text{KMnO}_4$ required was 24.6 ml.

1000 ml 1N $\text{KMnO}_4 \equiv 63.00 \text{ g oxalic acid.}$

$\therefore 1 \text{ ml } 0.819 \frac{N}{10} \text{KMnO}_4 \equiv (0.0063 \times 0.819) \text{ g oxalic acid.}$

$\therefore 24.6 \text{ ml } \frac{N}{10} \text{KMnO}_4 \equiv (0.0063 \times 24.6 \times 0.819) \text{ g oxalic acid} = 0.1269 \text{ g oxalic acid.}$

Again, $126.00 \text{ g oxalic acid} \equiv 88.00 \text{ g oxalate.}$

$\therefore 0.1269 \text{ g oxalic acid} \equiv \frac{88}{126} \times 0.1269 \text{ g oxalate} = 0.0889 \text{ g oxalate.}$

$\therefore \text{Percentage of oxalate content in the complex} = \frac{0.0889}{0.1710} \times 100 = 51.84\%.$

Actual percentage of oxalate content in the complex = $\frac{88 \times 3}{490} = 53.87\%.$

Potassium tris(oxalato)ferrate(III) trihydrate (**1**) is stable with 2(N) sulphuric acid in dark but dissociates quantitatively in light within 5–10 min into Fe(II)oxalate and carbon dioxide.



1.3.3 Estimation of crystallized water

The crystallized water in **1** was estimated gravimetrically. Accurately weighed sample (0.8875 g) was placed in an ordinary hot air oven, and the temperature was set at around 120°C. After repeating the process of heating, cooling, and weighing several times, constant weight appeared at 0.7950 g. Hence expelled water was (0.8875 – 0.7950) g = 0.0925 g, and percentage of crystallized water $\frac{0.0925}{0.8875} \times 100 = 10.42\%$.

Actual percentage of water in the compound = $\frac{54}{490} \times 100 = 11.02\%$.

2. Derivation of Empirical Formula

The analytical results obtained in the above experiments may be summarized as follows.

Percentage of Fe(III) = 13.12%

Percentage of oxalate = 51.84%

Percentage of water = 10.42%

Hence percentage of K(I) = $[100 - (13.12 + 51.84 + 10.42)]\% = 24.62\%$

Actual percentage of K(I) = $\frac{39.13}{490} \times 100 = 23.93\%$

Hence the molecular formula of the complex may be derived as follows. K(I) : Fe(III) : oxalate : H₂O = 24.62 : 13.12 : 51.84 : 10.42

Dividing each amount by their respective atomic mass or molecular (or formula) mass, hence having the ratio of their moles is

$$\begin{aligned} & \frac{24.62}{39.10} : \frac{13.12}{55.85} : \frac{51.84}{88.00} : \frac{10.42}{18.00} \\ &= 0.63 : 0.23 : 0.59 : 0.58 \\ &= \frac{0.63}{0.23} : \frac{0.23}{0.23} : \frac{0.59}{0.23} : \frac{0.58}{0.23} \\ &= 2.73 : 1.00 : 2.57 : 2.52 \end{aligned}$$

$$\approx 3 : 1 : 3 : 3$$



So the present experimental data propose the molecular formula of **1** as $\text{K}_3[\text{Fe}(\text{ox})_3] \cdot 3\text{H}_2\text{O}$ which matches exactly with the formula of **1**.

3. Conclusion

The present analytical results match fairly well with the molecular formula of potassium tris(oxalato)ferrate(III) trihydrate. Using this method, students can learn the approach for titrimetric analysis of the compound and determination of molecular formula therefrom.

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Suggested Reading

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